Project Progress - Week 1 of March 2025

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This is a file that contains the tracking of the activities related to the OFC project during the first week of March 2025. The activities are divided into groups and a summary of the progress is provided.

1 Initial Status of the Project

The project has made significant progress up until the first week of March 2025. The key achievements include:

- Successful implementation and validation of the optical force calculation function in the dipole approximation.
- Development of test and validation scripts that confirm the correct dependence on the permittivity.
- Advancement in the theoretical documentation, particularly in the section on polarizability.
- Detailed derivation of the quasiestatic polarizability formula for a dielectric sphere.

2 Potential Tasks for the Week

The following tasks have been identified as potential areas of focus for the first week of March 2025:

- Optimize the functions for querying the refractiveindex.info database.
- Complete the integration of permittivities into an specific equation.
- Finish documenting and providing examples of usage.

3 Week Tracking

3.1 Monday - March 3, 2025

The repository structure has been fixed and the management of Python modules has been addressed. Additionally, I have been learning how to compile LaTeX projects located inside the repository. This document serves as an example of the compilation process.

3.2 Tuesday - March 4, 2025

The tasks for the week have been specified and the progress in documentation, code, and testing has been tracked. We are going to focus on the algorithm used to obtain the polarizability of the particles. In the "dipole momenta" branch, a first version of ".h" and ".cpp" files has been created as a first approach to modular development of the functionalities. I have encountered some issues with warnings and errors flagged by the editor in VSCode. While the code works properly, the editor flags some issues that are not real. It may be related to the editor's settings or installation. Apart from this, it has been observed that a CMakefile must be created, an include directory must be added, and we need to explore how to include different functions without having to import multiple files.

4 Current Next Steps

The next steps for the project include:

- Development of a calculation method for Optical Forces given a polarizability, a field, and a field gradient.
- Documentation of quasiestatic polarizability theory.
- Refactor the code to improve modularity and readability.
- Docstring the first module and provide examples of usage.
- Implement the refractive index querying functions.
- Integrate the permittivities into the optical force calculation.
- Continue documenting the theoretical background and the code.